MIL-C-63371B(AR)
31 July 1984
SUPERSEDING
MIL-C-63371A(AR)
2 April 1981

MILITARY SPECIFICATION SHEET

CIRCUIT CARD ASSEMBLIES, MINE DISPENSER: M128 GENERAL SPECIFICATION FOR

This specification is approved for use by the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers quality assurance provisions and preparation for delivery for circuit card assemblies, hereinafter referred to as "assemblies", which form part of the Mine Dispenser, M128.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification For.
MIL-D-63303	-	Dispenser, Mine, Ground Vehicle: M128

STANDARDS

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-454	-	Standard General Requirements for Electronic Equipment.
MIL-STD-810	-	Environmental Test Methods for Aerospace and Ground Equipment.

FSC 1095

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (see 6.5)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER

9290227	-	Circuit Card Assembly No. 1
9291352	-	Circuit Card Assembly No. 2
9291224	-	Circuit Card Assembly No. 3
9290623	-	Circuit Card Assembly No. 4
9290647	-	Circuit Card Assembly No. 5 & 6
9290651	-	Circuit Card Assembly No. 7 & 8
9335623	-	Circuit Card Assembly No. 9 & 10
9290656	-	Circuit Card Assembly No. 11 & 12
9290661	-	Circuit Card Assembly No. 13
9291379	-	Circuit Card Assembly No. 14
9290793	-	Circuit Card Assembly No. 15
9290669	-	Circuit Card Assembly No. 16
9291348	-	Circuit Card Assembly No. 17
P9290227	-	Circuit Card Assembly No. 1
P9291352	-	Circuit Card Assembly No. 2
P9291224	-	Circuit Card Assembly No. 3
P9290623	-	Circuit Card Assembly No. 4
P9290647	-	Circuit Card Assembly No. 5 & 6
P9290651	-	Circuit Card Assembly No. 7 & 8
P9335623	-	Circuit Card Assembly No. 9 & 10
P9290656	-	Circuit Card Assembly No. 11 & 12
P9290661	-	Circuit Card Assembly No. 13
P9291379	-	Circuit Card Assembly No. 14
P9290793	-	Circuit Card Assembly No. 15
P9290669	-	Circuit Card Assembly No. 16
P9291348	-	Circuit Card Assembly No. 17

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings, specifications and standards.

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3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on the applicable drawings, and with all requirements specified in applicable specifications and standards.

3.3 Burn-in. Prior to submittal for acceptance testing each assembly will be conditioned for a minimum of twelve (12) hours at 52 ± 2°C (125°F Ref.) with B+ power applied.

3.4 Transportation vibration. Assemblies shall be capable of meeting the requirements of 3.7 after being tested as specified in 4.5.2.

3.5 High temperature. Circuit card assemblies shall be capable of withstanding storage at 71°C (160°F) and shall be fully operational at 52°C (125°F) when tested as specified in 4.5.3.

3.6 Low temperature. Circuit card assemblies shall be capable of withstanding storage at -57°C (-70°F) and shall be fully operational at -32°C (-25°F) when tested as specified in 4.5.4.

3.7 Electrical function. The electrical function requirements for each circuit card assembly are defined on the specification sheets as tabulated in Table 1.

TABLE 1

Circuit Card Assembly Specification Sheets

Drawing No.	Assembly No.	Slash No.
9290227	1	/1
9291352	2	/2
9291224	3	/3
9290623	4	/4
9290647	5&6	/5
9290651	7&8	/6
9335623	9&10	/7
9290656	11&12	/8
9290661	13	/9
9291379	14	/10
9290793	15	/11
9290669	16	/12
9291348	17	/13

3.8 Soldering. All soldered connections shall comply with the requirements of MIL-STD-454, Requirement 5.

3.9 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

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3.10 Workmanship. All assemblies shall be fabricated and finished in a thorough workmanlike manner and all manufacturing, processing and assembly operations shall be correctly performed. Assemblies shall be clean and free of burrs, sharp edges, surface defects, flux, solder bridges or splashes, lifted pads, grease or other foreign matter. The soldered connections shall have a smooth bright appearance with metallic luster and shall not have a chalky, gritty or irregular surface, nor exhibit protrusions, pits, or voids which expose base metal or where the bottom of the pit or void is not visible. Cleaning methods used shall not contaminate or degrade any component or assembly. All required marking shall be neat and sharply defined. Packing materials shall be dry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection (see 4.3).

b. Quality Conformance Inspection (see 4.4).

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of parts and assemblies in the quantities indicated in Table 2. The first article test need only be conducted when assemblies are purchased as repair parts. When purchased as part of a complete dispenser system, first article will be conducted in accordance with MIL-D-63303(AR).

4.3.2 Inspections to be performed. See MIL-A-48078 and Table 2 specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE 2 - First article inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-C-63371B(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
	Circuit Card Assemblies		All - See Table 1
			NEXT HIGHER ASSEMBLY 9290236

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Burn-in	10 ea	100%	3.3	4.5.1
	Examination for defects	10 ea	100%	3.2	4.4.2.1
	Transportation vibration	10 ea	100%	3.4	4.5.2
	High temperature	10 ea	100%	3.5	4.5.3
	Low temperature	10 ea	100%	3.6	4.5.4

NOTE: A total of 120 circuit card assemblies are required for First Article Testing (10 each of 12 part numbers).

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification for Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63371B(AR)

PARAGRAPH 4.4.2.1	TITLE Circuit Card Assembly		SHEET 1 OF 1		DRAWING NUMBER All assemblies - See Table 1
					NEXT HIGHER ASSEMBLY 9290236
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any component missing, damaged, or incorrect		0.40%	3.2	Visual
102	Any component mounted incorrectly or with reversed polarity		0.40%	3.2	Visual
103	Soldering incorrect or inadequate		0.40%	3.8	Visual
104	Insulating compound missing or incorrect or covering contact surface		0.40%	3.2	Visual
105	Burn-in		100%	3.3	4.5.1
106	Electrical function		100%	3.7	4.5.5
Minor					
201	Part number missing, incorrect or illegible		0.65%	3.2	Visual
202	Poor workmanship		1.00%	3.10	Visual

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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4.4.3 Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and section 6.3 herein.

4.5 Methods of inspection.

4.5.1 Burn-in. Using equipment in accordance with 4.4.4, each assembly shall be subjected to a minimum of twelve hours of powered operation while being maintained at 52 $\pm$ 2°C.

4.5.2 Transportation vibration. Assemblies selected for this test shall be mounted in a Card Rack Assembly, 9290236, or approved test fixture (see 4.4.4) which simulated the Card Rack. The fixture shall then be tested in accordance with the procedures specified in MIL-STD-810, Test 514.2, Procedure VIII, Curve Y.

4.5.3 High temperature. Assemblies selected for this test shall be mounted in a suitable fixture (see 4.4.4) and tested to the requirements of 3.5 in accordance with Procedure I of Test 501.1, MIL-STD-810.

4.5.4 Low temperature. Assemblies selected for this test shall be mounted in a suitable fixture (see 4.4.4) and tested to the requirements of 3.6 in accordance with Procedure I of Test 502.1, MIL-STD-810.

4.5.5 Electrical function. For electrical characteristics, refer to applicable specification sheet as tabulated in Table I.

5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with the packaging data sheets listed in Table 3.

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TABLE III. PACKAGING DATA SHEETS & CCA NUMBERS

Packaging Data Sheet No.	Circuit Card Assembly No.
P9290227	1
P9290618	2
P9291224	3
P9290623	4
P9290647	5&6
P9290651	7&8
P9290652	9&10
P9290656	11&12
P9290661	13
P9290665	14
P9290793	15
P9290669	16
P9291348	17

6. NOTES

6.1 Intended use. The circuit card assemblies referred to in this document form a part of the M128 Mine Dispenser; one of the system in the family of Scatterable Mines.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Provisions for submission of first article samples.

c. Provisions of MIL-A-48078.

6.3 Submission of inspection equipment for design approvals. See MIL-A-48078. Submit designs as required to: Commander, US Army Armament Research and Development Center, ATTN: DRSMC-QAT-I, Dover, NJ 07801.

* 6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by and identified as US Army Armament Research and Development Command (ARRADCOM), Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDC.

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* 6.5 Changes from previous issue. The margins of this specification are marked (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-63371B (AR)	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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